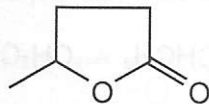


- 29 γ -Valerolactone is a naturally occurring organic compound found in fruits and could be a possible bio-fuel alternative to ethanol.



γ -valerolactone

Which statements about this compound are correct?

- 1 It can be prepared by warming 4-hydroxypentanoic acid, $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_2\text{CO}_2\text{H}$, in the presence of an acid catalyst.
- 2 It is an ester.
- 3 It reacts readily with warm, aqueous alkali.

A 1, 2 and 3 **B** 1 and 3 only **C** 2 and 3 only **D** 2 only